

Kinetic Modeling of Cell Signaling

Module 1: Reaction networks as dynamical systems

Jim Faeder

MSCBIO 2025 · Introduction to Bioinformatics Programming in Python

Fall 2026

Resources

- ▶ **BioNetGen:** software, documentation, and tutorials: bionetgen.org
- ▶ **VS Code extension:** search the Extensions panel for *BioNetGen*
- ▶ **bngsim:** fast ODE and stochastic simulation of BioNetGen models from Python: bngsim.readthedocs.io
Install with `pip install bngsim bionetgen`
- ▶ **Course repository:** github.com/jrfaeder/RBM_Intro_BioNetGen
 - `models/` BNGL models used in the slides
 - `notebooks/` Jupyter notebooks for simulation and fitting
 - `Slides/pdf/` slides and handouts



Course repository
models, notebooks, slides

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4. encode the same network in the **BioNetGen language** (BNGL), simulate it, and interpret the results
5. explain how a **negative feedback** loop shapes the dynamics of a signaling response

Part 1

Why model reaction networks?

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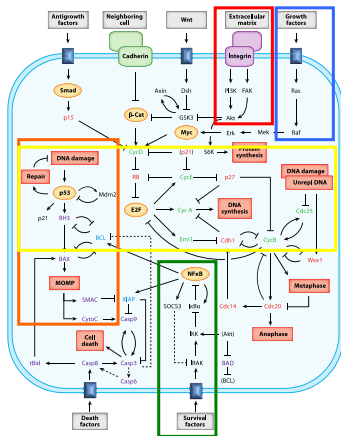
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What a model gives us

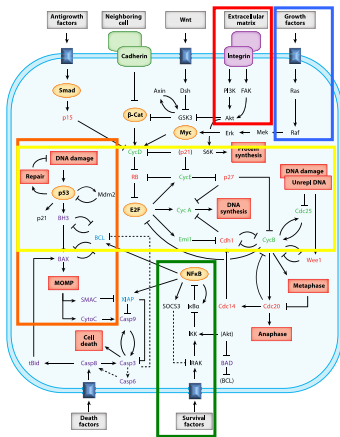
- ▶ A precise statement of a mechanism
- ▶ Predictions of dynamics we cannot easily measure
- ▶ A way to test whether a mechanism can explain data
- ▶ Hypotheses for the next experiment

Toward an integrated model of the cell



Cell fate is decided by many interacting signaling modules: growth-factor signaling, the cell cycle, DNA damage, apoptosis.

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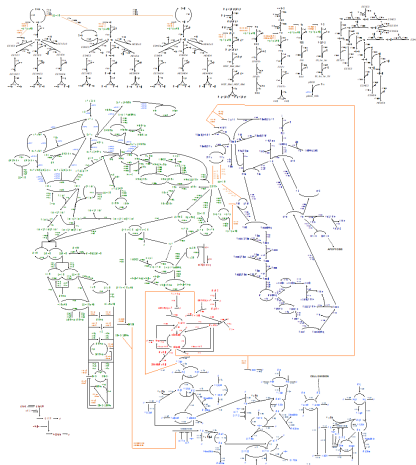
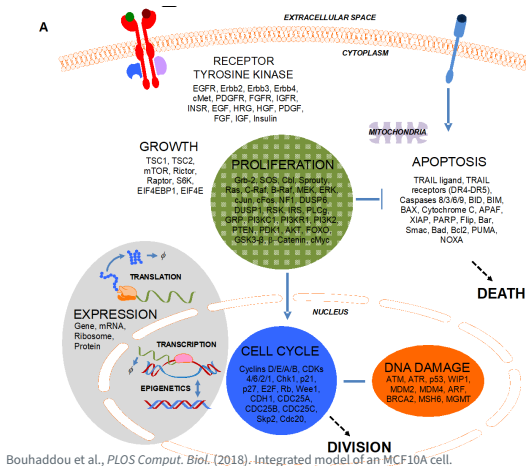


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How do we link these modules together into models that predict how cells respond?

Tyson and Novak, *Annu. Rev. Phys. Chem.* **61**, 219–240 (2010)

Large models are built from simple reaction steps

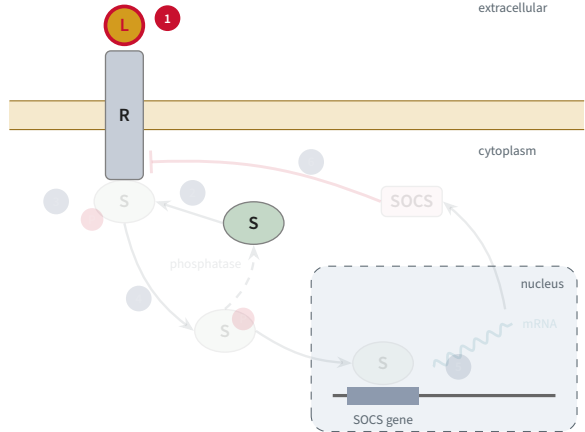


Part 2

A cytokine signaling circuit

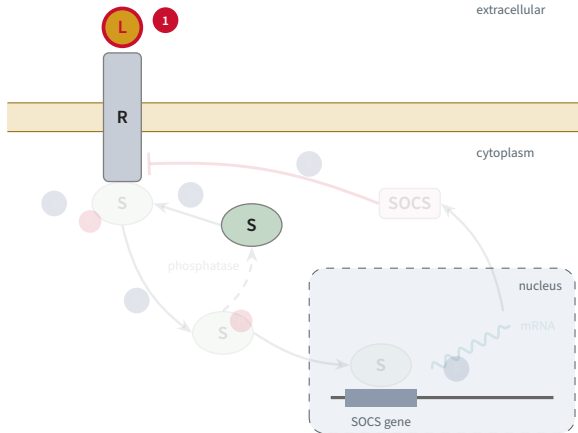
Cytokines signal through receptors, STATs, and SOCS feedback

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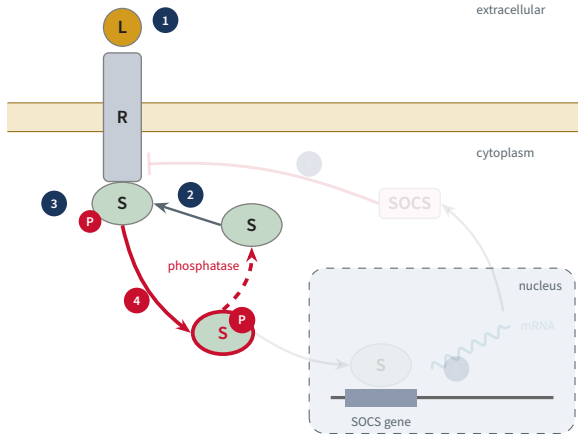
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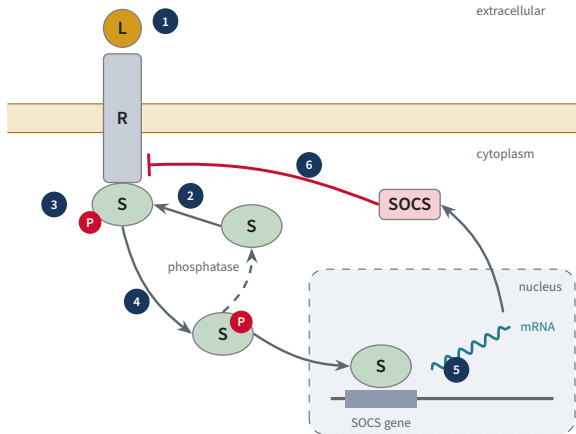
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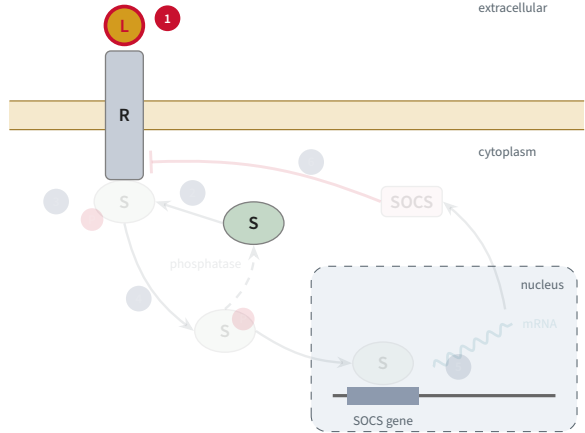
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- ▶ Binding to their **receptor** activates receptor-associated JAK kinases.
- ▶ **STAT** transcription factors dock on the active receptor, are phosphorylated, and move to the nucleus.
- ▶ pSTAT induces genes including **SOCS** (suppressor of cytokine signaling).
- ▶ SOCS binds the receptor and **shuts off further STAT activation**: negative feedback.



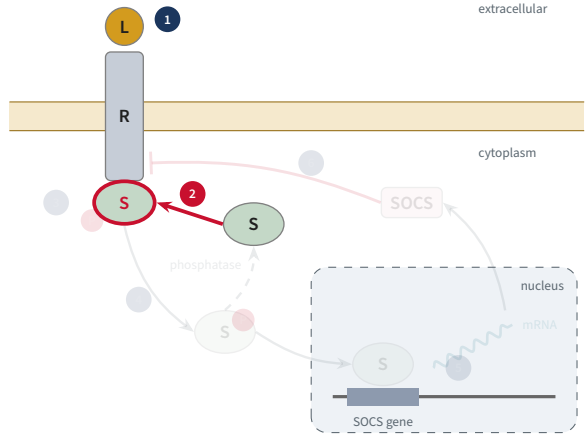
We will build the model one reaction step at a time

1. Ligand-receptor binding



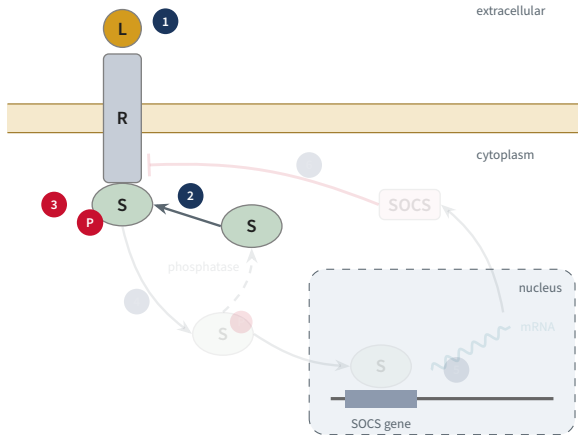
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2. STAT binds the ligand-bound receptor



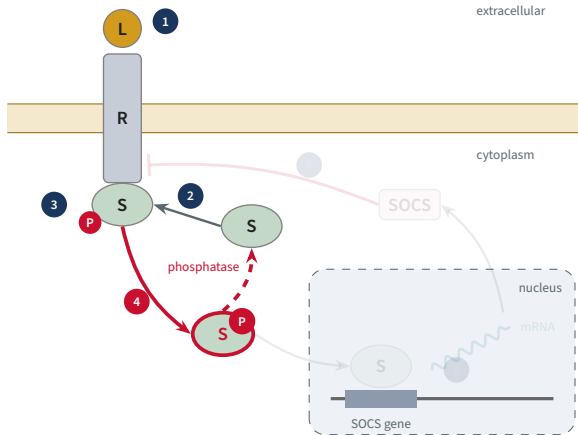
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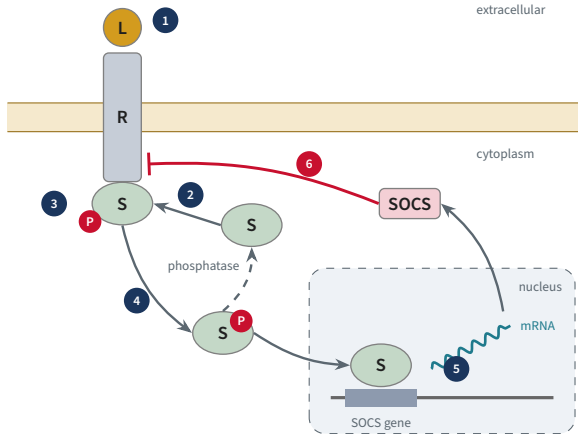
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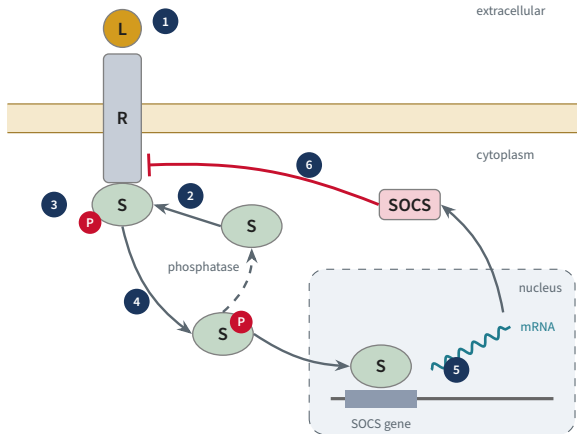
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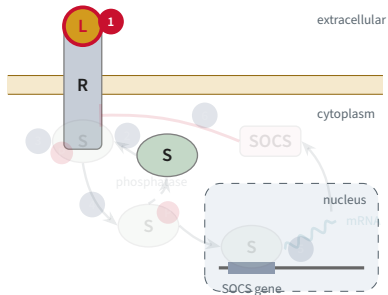
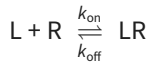
Each step adds reactions to the previous model.
The files are `models/stat_step1_binding.bngl`
through `stat_step6_feedback.bngl`.



Part 3

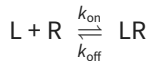
Step 1: Ligand–receptor binding

Binding is a reversible pair of reactions



Units: k_{on} in $\mu\text{M}^{-1}\text{min}^{-1}$, k_{off} in min^{-1}

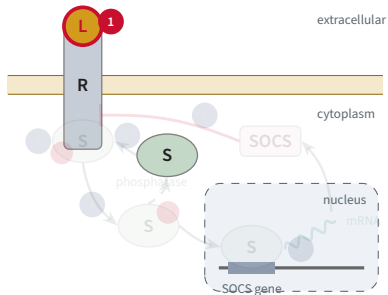
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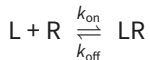
binding: $v_1 = k_{\text{on}} [L] [R]$

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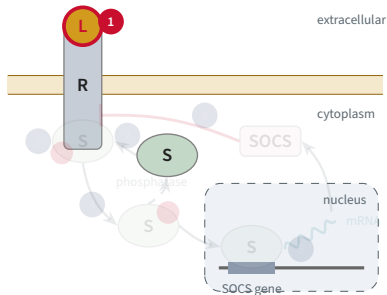
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Law of mass action

The rate of an elementary reaction is proportional to the product of the concentrations of its reactants. The proportionality constant is the **rate constant**.



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Each reaction contributes to the rate of change of its species

	v_1 (bind)	v_2 (unbind)
L	-1	+1
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Reactants are consumed ($-$), products are created ($+$).

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Ordinary differential equations (ODEs) follow mechanically from the list of reactions and their rate laws.

The stoichiometry matrix gives a compact general form

Collect the concentrations in a vector $\mathbf{X}$ and the reaction rates in $\mathbf{v}$:

$$\frac{d\mathbf{X}}{dt} = \mathbf{S} \mathbf{v}(\mathbf{X}), \quad \frac{d}{dt} \begin{pmatrix} [L] \\ [R] \\ [LR] \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & +1 \\ -1 & +1 \\ +1 & -1 \end{pmatrix}}_{\mathbf{S}} \underbrace{\begin{pmatrix} k_{\text{on}}[L][R] \\ k_{\text{off}}[LR] \end{pmatrix}}_{\mathbf{v}(\mathbf{X})}$$

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- ▶ $\mathbf{S}$ has one row per species and one column per reaction.
- ▶ This form holds for *any* reaction network; only $\mathbf{S}$ and $\mathbf{v}$ change.

Conservation laws and equilibrium

Conservation. Binding only moves molecules between free and bound forms:

$$[L] + [LR] = L_0$$

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Small models can be solved by hand. Real signaling networks cannot, so we solve the ODEs numerically.

ODEs for reaction networks are solved numerically

Simplest idea (forward Euler): take small time steps

Δt ,

$$\mathbf{X}(t + \Delta t) \approx \mathbf{X}(t) + \Delta t \mathbf{S} \mathbf{v}(\mathbf{X}(t))$$

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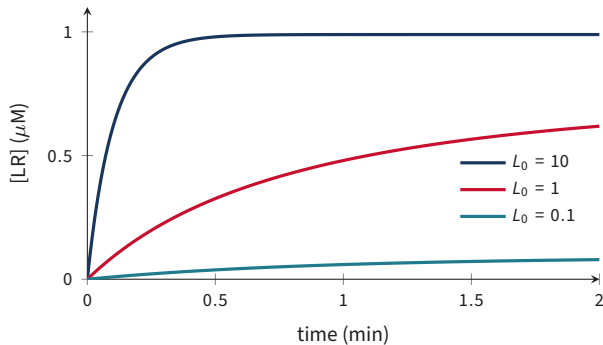
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In BioNetGen

```
simulate({method=>"ode", ...})  
uses CVODE.
```

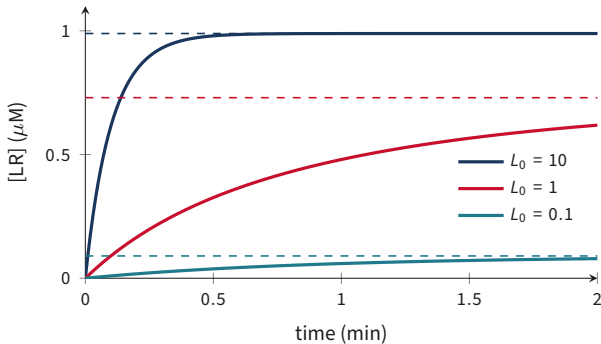
In Python, `model.setup_simulator()` returns a `libRoadRunner` simulator for the same model.

Binding approaches equilibrium faster and further at higher ligand



$$R_0 = 1 \mu\text{M}, k_{\text{on}} = 1 \mu\text{M}^{-1}\text{min}^{-1}, \\ k_{\text{off}} = 0.1 \text{ min}^{-1} (K_D = 0.1 \mu\text{M}).$$

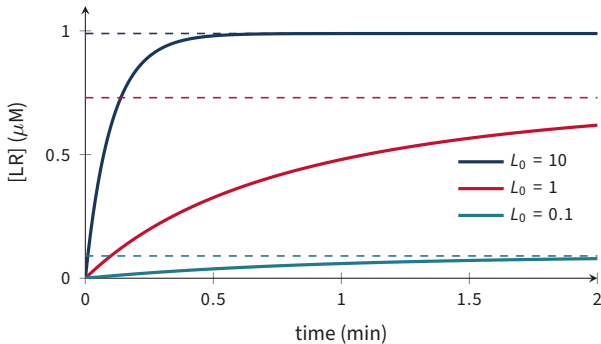
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The relaxation rate is $k_{\text{on}}([L] + [R]) + k_{\text{off}}$ near equilibrium, so higher ligand also means a faster response.

The same model in BNGL

```
begin model

begin parameters
  L0      10      # ligand (uM)
  R0      1       # receptor (uM)
  kon     1       # binding (1/uM/min)
  koff    0.1     # unbinding (1/min)
end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
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  Molecules Lfree  L
  Molecules Rfree  R
  Molecules LR     LR
end observables

begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <=> LR kon, koff
end reaction rules

end model

generate_network({overwrite=>1})
simulate({method=>"ode", t_end=>2, n_steps=>200})
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- ▶ **reaction rules:** $\leftrightarrow$ is reversible; forward and reverse rate constants

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  Molecules LR     LR
end observables

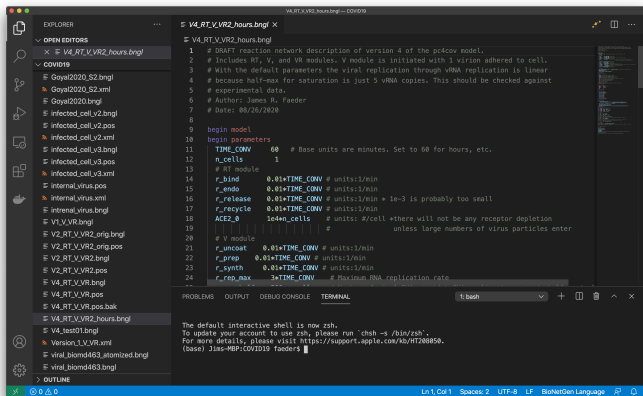
begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <-> LR kon, koff
end reaction rules

end model

generate_network({overwrite=>1})
simulate({method=>"ode", t_end=>2, n_steps=>200})
```

- ▶ **observables:** quantities to output
- ▶ **reaction rules:** $\leftrightarrow$ is reversible; forward and reverse rate constants
- ▶ **actions** after `end model`: generate the network, then simulate

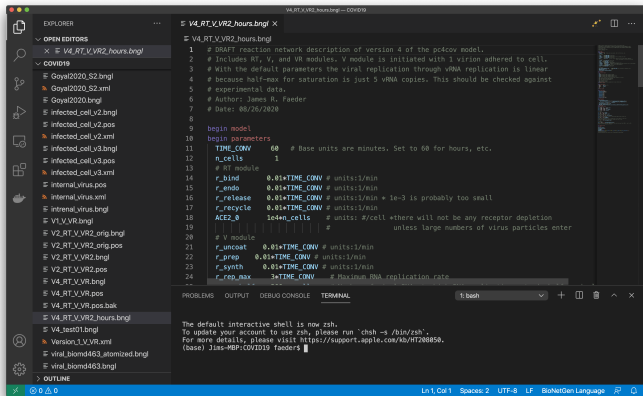
Running a model in VS Code



► Open a .bngl file

VS Code extension: Ali Sinan Saglam

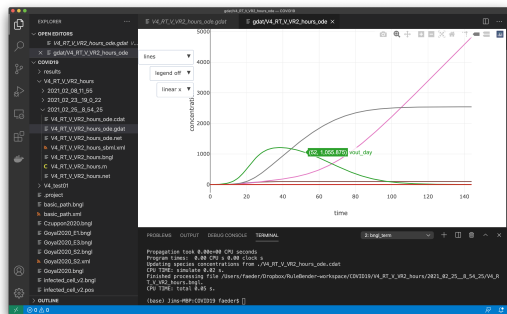
Running a model in VS Code



- ▶ Open a .bngl file
- ▶ Click the **run** button in the editor toolbar
- ▶ Results go to
BNG-results/<model>/
<timestamp>/

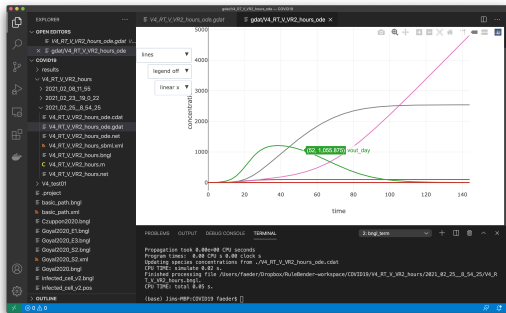
VS Code extension: Ali Sinan Saglam

Interactive plotting and Python notebooks

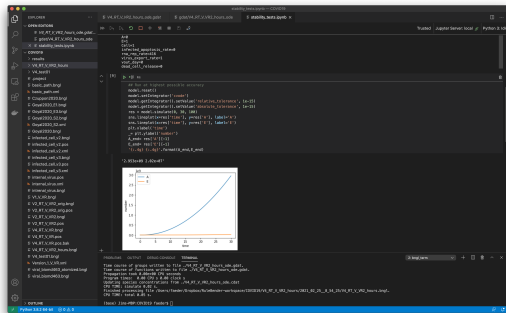


Built-in plotting of .gdat output

Interactive plotting and Python notebooks



Built-in plotting of .gdat output

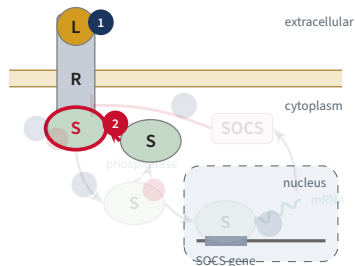


Deeper analysis in Jupyter with PyBioNetGen + libRoadRunner

Part 4

Building the signaling circuit in BNGL

Step 2: STAT binds the ligand-bound receptor



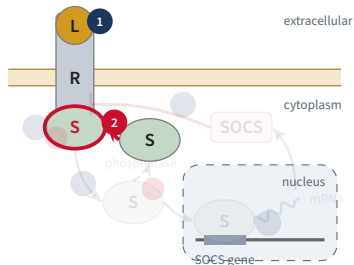
STAT can bind *only* the ligand-bound receptor LR, never free R. This is how ligand binding **enables** STAT recruitment.

Step 2: STAT binds the ligand-bound receptor



New lines in `stat_step2_recruit.bngl`:

```
begin parameters
  S0      10    # STAT (uM)
  kSon    1     # STAT binding (1/uM/min)
  kSoff   1     # STAT unbinding (1/min)
end parameters
begin molecule types
  S
  LRS
end molecule types
begin species
  S S0
end species
begin reaction rules
  Sbind: LR + S <-> LRS kSon, kSoff
end reaction rules
```



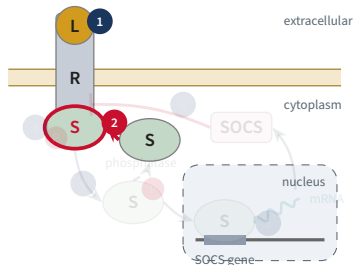
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New lines in `stat_step2_recruit.bngl`:

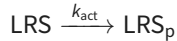
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begin parameters
  S0      10    # STAT (uM)
  kSon    1     # STAT binding (1/uM/min)
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end parameters
begin molecule types
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begin reaction rules
  Sbind: LR + S <-> LRS kSon, kSoff
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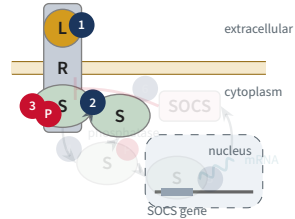
STAT can bind *only* the ligand-bound receptor LR, never free R. This is how ligand binding **enables** STAT recruitment.

Every new species needs a molecule type; species present at $t = 0$ also need an initial amount.

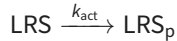
Step 3: receptor-bound STAT is activated



A receptor-associated kinase (JAK) phosphorylates the docked STAT: a **first-order**, irreversible reaction with $v = k_{\text{act}}[\text{LRS}]$.



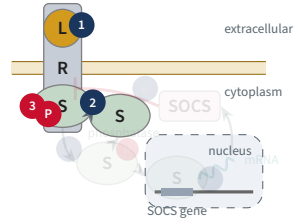
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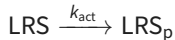
A receptor-associated kinase (JAK) phosphorylates the docked STAT: a **first-order**, irreversible reaction with $v = k_{\text{act}}[\text{LRS}]$.

```
kact 1    # STAT activation (1/min)
LRSp
Sact: LRS -> LRSp kact
```

one new line each in parameters, molecule types, and reaction rules



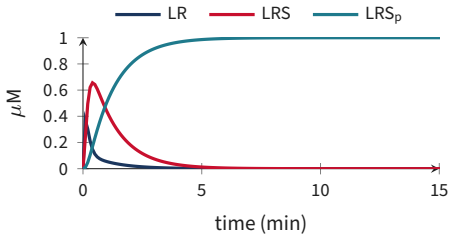
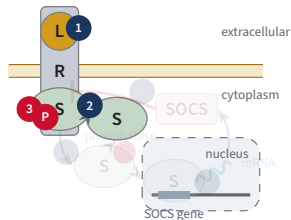
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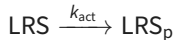
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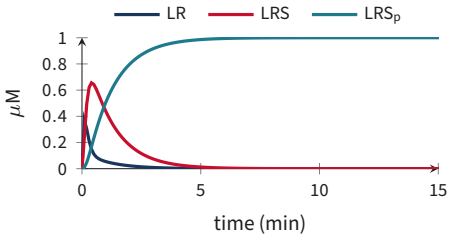
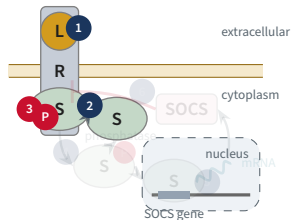


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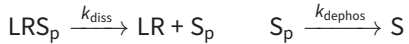
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LRSp
Sact: LRS -> LRSp kact
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one new line each in parameters, molecule types, and reaction rules

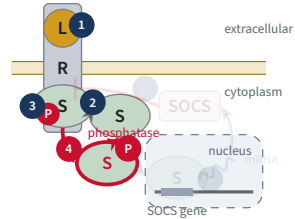
Problem: pSTAT never leaves the receptor. Every receptor ends up stuck as LRS_p.



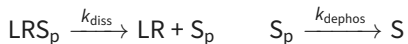
Step 4: activated STAT dissociates from the receptor



Releasing pSTAT frees the receptor to activate another STAT. A phosphatase returns free pSTAT to the inactive pool.

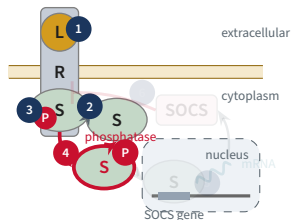


Step 4: activated STAT dissociates from the receptor

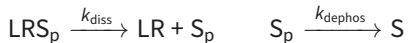


Releasing pSTAT frees the receptor to activate another STAT. A phosphatase returns free pSTAT to the inactive pool.

```
k diss    1    # pSTAT release (1/min)
k dephos  0.1  # dephosphorylation (1/min)
Sp
S release: LRSp -> LR + Sp k diss
S dephos:  Sp -> S k dephos
```

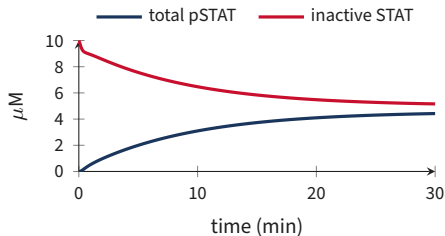
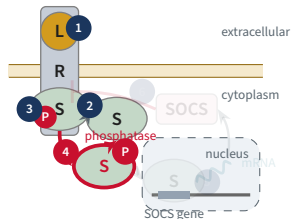


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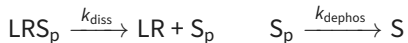


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Srelease: LRSp -> LR + Sp kdiss
Sdephos:  Sp -> S kdephos
```



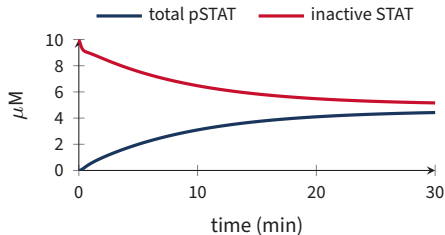
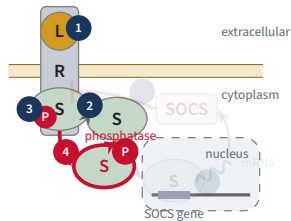
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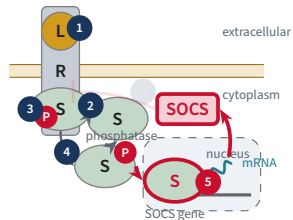
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Sdephos:  Sp -> S kdephos
```

pSTAT now reaches a steady state set by the balance of activation and dephosphorylation.



Step 5: pSTAT induces SOCS expression

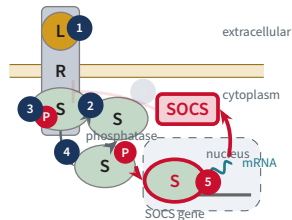
```
# pSTAT binds the SOCS promoter
Gbind: Sp + G <-> GSp kGon, kGoff
# Bound promoter is transcribed
Transcribe: GSp -> GSp + mSOCS ktx
# mRNA degradation
mRNAdeg: mSOCS -> 0 kmdeg
# Translation
Translate: mSOCS -> mSOCS + SOCS kt1
# SOCS protein degradation
SOCSdeg: SOCS -> 0 kSOCSdeg
```



Step 5: pSTAT induces SOCS expression

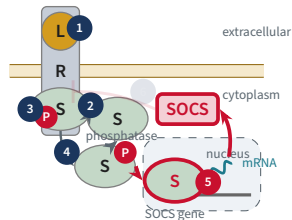
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- **Synthesis:** a species on both sides (GSp, mSOCS) is a template and is not consumed.



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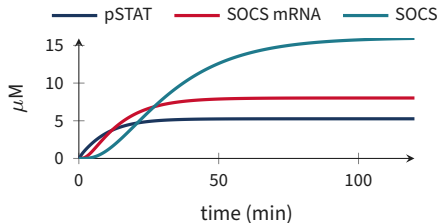
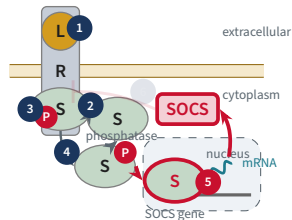


- **Synthesis:** a species on both sides (GSp , $mSOCS$) is a template and is not consumed.
- **Degradation:** $\emptyset$ means nothing; the molecule is removed.

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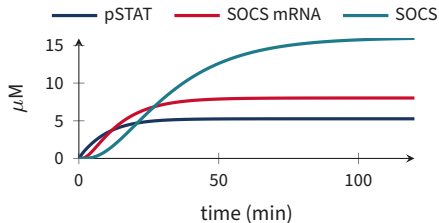
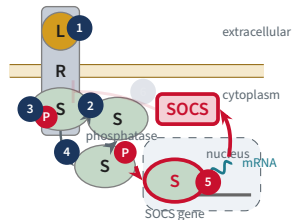


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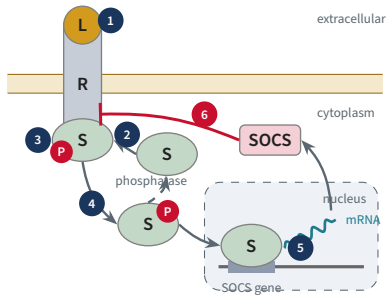
SOCS accumulates, but it has no target yet, so pSTAT is unaffected.



Step 6: SOCS binds the receptor and blocks STAT recruitment



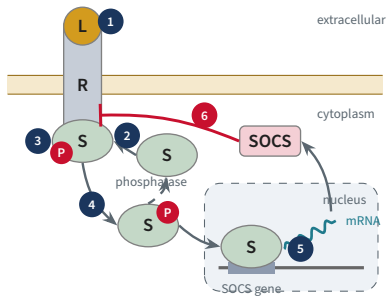
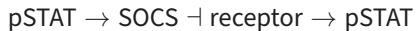
SOCS competes with STAT for the active receptor. LR·SOCS cannot recruit STAT, so pSTAT production drops.



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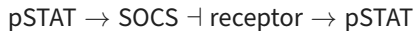
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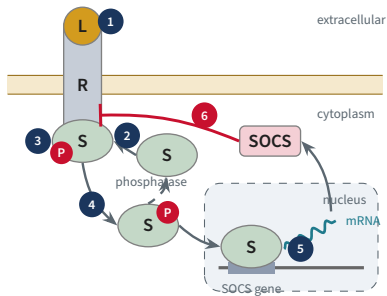
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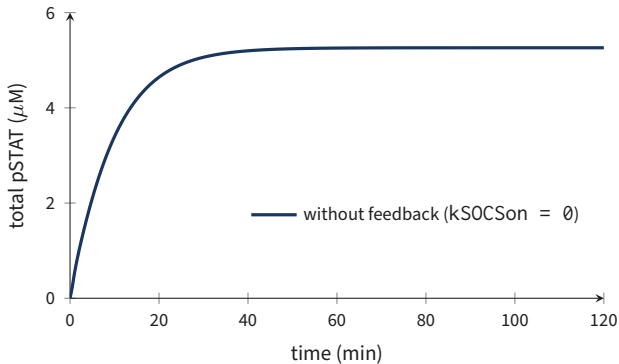
```

kSOCSon      1      # (1/uM/min)
kSOCSoff     0.01   # (1/min)
LRSOCS
SOCSbind: LR + SOCS <=> LRSOCS kSOCSon, kSOCSoff

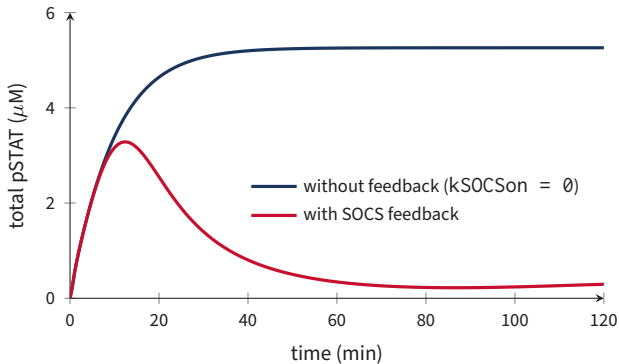
```



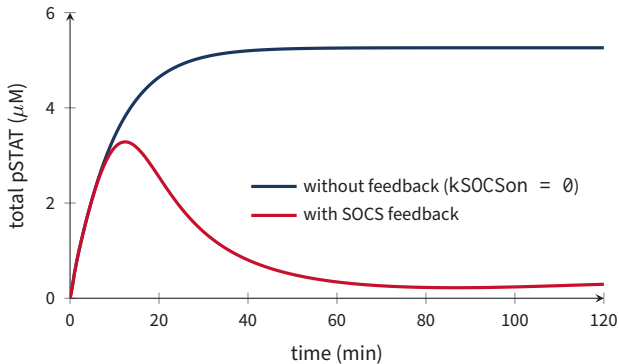
Negative feedback turns a sustained signal into a pulse



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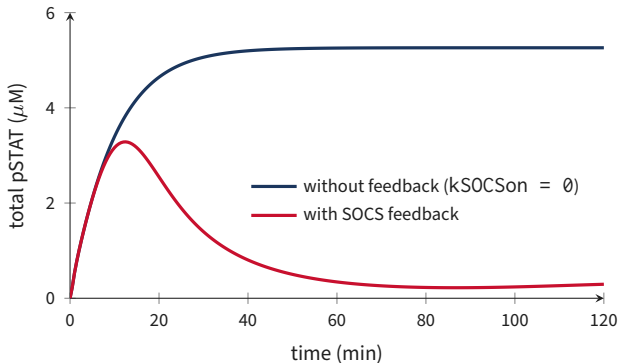


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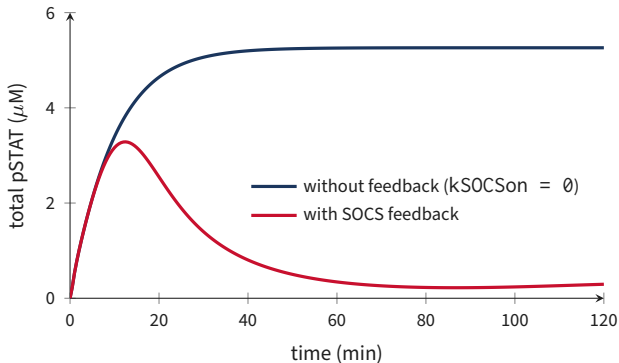
- Early response is identical: SOCS has not been made yet.

Negative feedback turns a sustained signal into a pulse



- ▶ Early response is identical: SOCS has not been made yet.
- ▶ As SOCS accumulates it sequesters receptors, and pSTAT falls to a low level despite continued ligand.

Negative feedback turns a sustained signal into a pulse



- ▶ Early response is identical: SOCS has not been made yet.
- ▶ As SOCS accumulates it sequesters receptors, and pSTAT falls to a low level despite continued ligand.
- ▶ The delay comes from transcription and translation.

Both curves come from `stat_step6_feedback.bngl`, which runs the model twice using `setParameter`.

The complete network

	reaction	rate
1	$L + R \rightleftharpoons LR$	k_{on}, k_{off}
2	$LR + S \rightleftharpoons LRS$	k_{Son}, k_{Soff}
3	$LRS \rightarrow LRS_p$	k_{act}
4	$LRS_p \rightarrow LR + S_p$	k_{diss}
	$S_p \rightarrow S$	k_{dephos}
5	$S_p + G \rightleftharpoons GS_p$	k_{Gon}, k_{Goff}
	$GS_p \rightarrow GS_p + mSOCS$	k_{tx}
	$mSOCS \rightarrow 0$	k_{mdeg}
	$mSOCS \rightarrow mSOCS + SOCS$	k_{tl}
	$SOCS \rightarrow 0$	$k_{SOCSdeg}$
6	$LR + SOCS \rightleftharpoons LR \cdot SOCS$	$k_{SOCSon}, k_{SOCSoff}$

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- 11 lines of BNGL generate **15 reactions** (4 are reversible) among **12 species**.

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- ▶ 11 lines of BNGL generate **15 reactions** (4 are reversible) among **12 species**.
- ▶ **S** is a 12×15 matrix; BioNetGen builds it and the ODEs for you (see the .net file).
- ▶ Every species had to be named by hand: L, LR, LRS, LRS_p, LR·SOCS, ...

Part 5

Looking ahead

Naming every complex by hand does not scale

Our model made strong simplifying assumptions. Relax a few of them:

- ▶ SOCS can also bind receptors that already carry STAT (LRS, LRS_p)

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Each feature **multiplies** the number of distinct complexes, and each complex needs its own species name and its own copy of every reaction it takes part in.

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Our model made strong simplifying assumptions. Relax a few of them:

- ▶ SOCS can also bind receptors that already carry STAT (LRS, LRS_p)
- ▶ Ligand can dissociate from *any* receptor complex, not only LR
- ▶ The receptor has two chains, each recruiting its own STAT
- ▶ STATs dimerize before entering the nucleus

Each feature **multiplies** the number of distinct complexes, and each complex needs its own species name and its own copy of every reaction it takes part in.

Module 2: represent molecules with internal structure (binding sites and states) and write rules that generate the network automatically.

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We return to stochastic simulation in a later module.

Try it yourself

Using the models in `models/`:

1. Run `stat_step1_binding.bngl` at several values of L_0 and check the equilibrium against the formula for $[LR]_{eq}$.
2. In `stat_step4_release.bngl`, set `kdephos` to 0. What happens to pSTAT, and why?
3. In `stat_step6_feedback.bngl`, vary `kt1` (translation rate). How do the peak and the late level of pSTAT change?
4. Add a reaction that lets SOCS bind LRS as well as LR. What new species do you need?

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- ▶ Building the cytokine–STAT–SOCS circuit one step at a time shows how each mechanism contributes; negative feedback produces an adaptive pulse of pSTAT.
- ▶ Writing every complex as a separate species quickly becomes unmanageable. This motivates rule-based modeling, the subject of Module 2.